

LimitIQ

Dynamic Credit Line Management & Exposure Optimization

Executive decision brief

LimitIQ evaluates current limit, +10%, +20% and +30% candidates, then selects the highest eligible simulated risk-adjusted contribution under loss, exposure, payment-history, overextension and profitability controls. Deteriorating accounts are frozen or referred rather than automatically decreased.

Evidence class	Result
Observed source	30,000 Taiwan credit-card accounts; April–September 2005 behavior
Model estimate	Untouched-test ROC-AUC 0.781; Brier 0.133
Simulation	1,904 increases; TWD 13,925,685 contribution

Portfolio simulation

Accounts	6,000
Current / proposed limits	TWD 1,011,790,000 / TWD 1,096,231,000
Current / proposed expected loss	TWD 101,772,493 / TWD 105,871,843
Simulated incremental contribution	TWD 13,925,685
Simulated risk-adjusted return	21.95%
Early-warning / manual review	1,662

Model governance

Champion: Histogram gradient boosting. The model type and threshold (0.149) were selected using validation data before one untouched-test evaluation. Sex, education, marital status, age, customer ID and target are excluded from decisioning.

Boundaries and committee actions

All line response, drawdown and financial effects are deterministic simulations. PD is not causal. A production build needs current local-law review, verified ability-to-pay inputs, outcome monitoring, model validation, controlled overrides, change approval and rollback. The displayed management ECL is not an IFRS 9 allowance or regulatory-capital calculation.

LimitIQ is an educational portfolio demonstration using public and synthetic data. It is not a production credit-decision system and must not be used to make real lending decisions.